Impact of Major Events on GDP: A Data-Driven Analysis

CS 4331 – Applied Data Science

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Introduction

Large scale events like the Olympics, World Cup, Major Concerts and more attract attention and often have a huge impact on the economy of the place it is hosted. Big events usually lead to an increase in tourism, employment rates and new infrastructure.

Our project aims to explore and quantify how hosting these events affects a country's or city's economic growth.

Background

- Large-scale events such as the Olympics, FIFA World Cup, Formula 1, and major music festivals or concerts are often promoted as engines for economic growth and international visibility.
- Host cities invest heavily in infrastructure, tourism promotion, and event operations, expecting returns through GDP growth, job creation, and urban renewal.
- Governments and organizers often justify hosting by projecting:
  - Increases in GDP and foreign investment
  - Growth in tourism, construction, and service sectors
  - Creation of short-term and long-term employment opportunities
- Research shows short-term spikes in GDP due to visitor spending and temporary jobs, but long-term benefits are less consistent.
- Some hosts face debt burdens and underused infrastructure after events end.

Research Question

Primary Question

Do major international events such as the Olympics, World Cup, Formula1, and concerts lead to measurable changes in the GDP of host cities or countries before and after the event?

Secondary Questions

1. Is the impact short-term or long-term?
2. Does the type of event have a stronger bearing on the impact ?

Data Collection

- Our Dataset: Growth dataset Olympic Games and Football World Cup.xlsx
- The dataset includes: Host country and event year , Type of mega-event (Olympics or World Cup) , Economic indicators (GDP growth, tourism numbers, etc.) , Event-related finances , Cost of venues , Cost of organization , Revenue , Additional event metrics depending on the year
- Sources: https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/CPQEHN
- Challenges Faces: Inconsistent formatting , Non-uniform coverage , Difficulty verifying financial accuracy , and Different data types

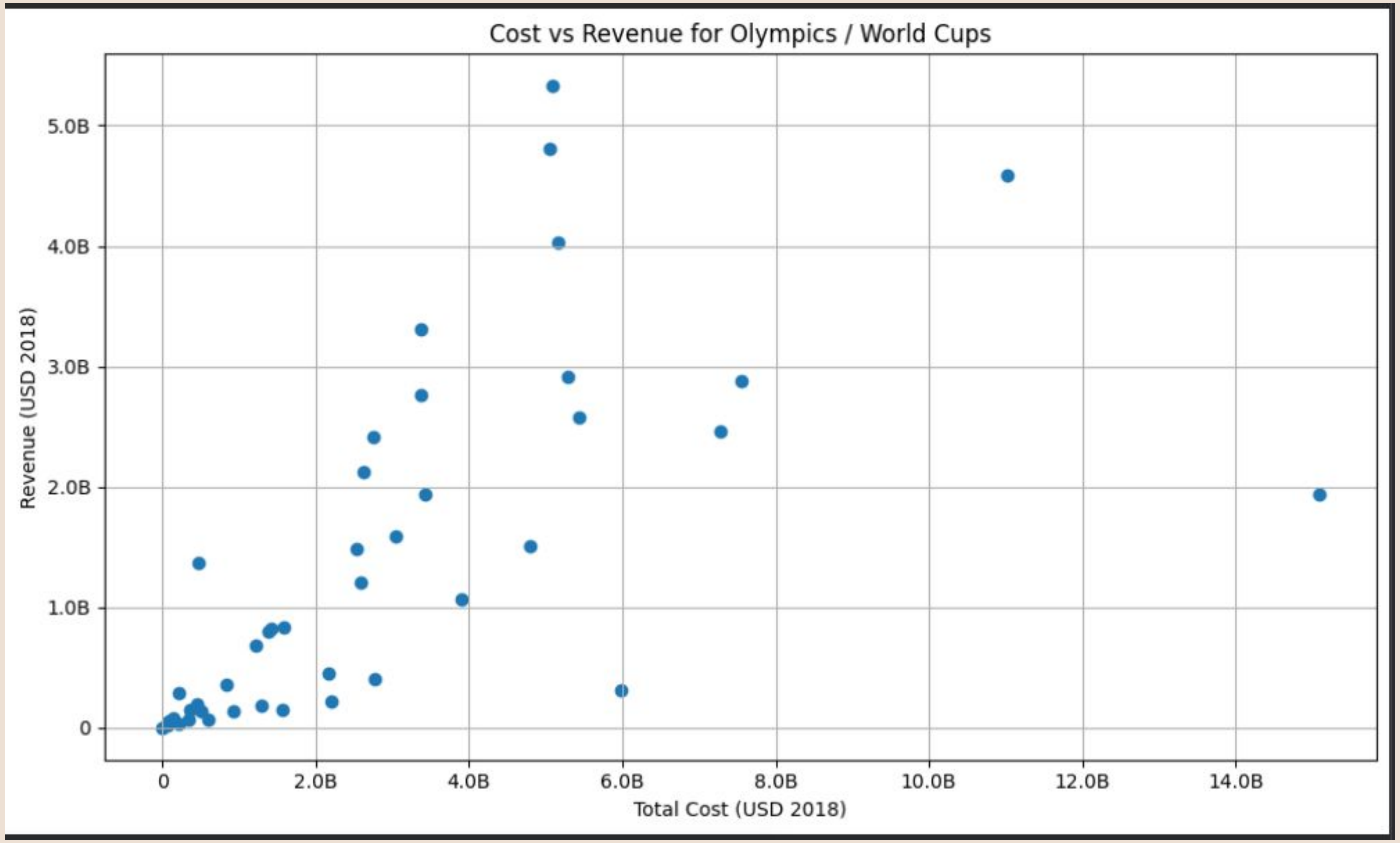
Data Cleaning

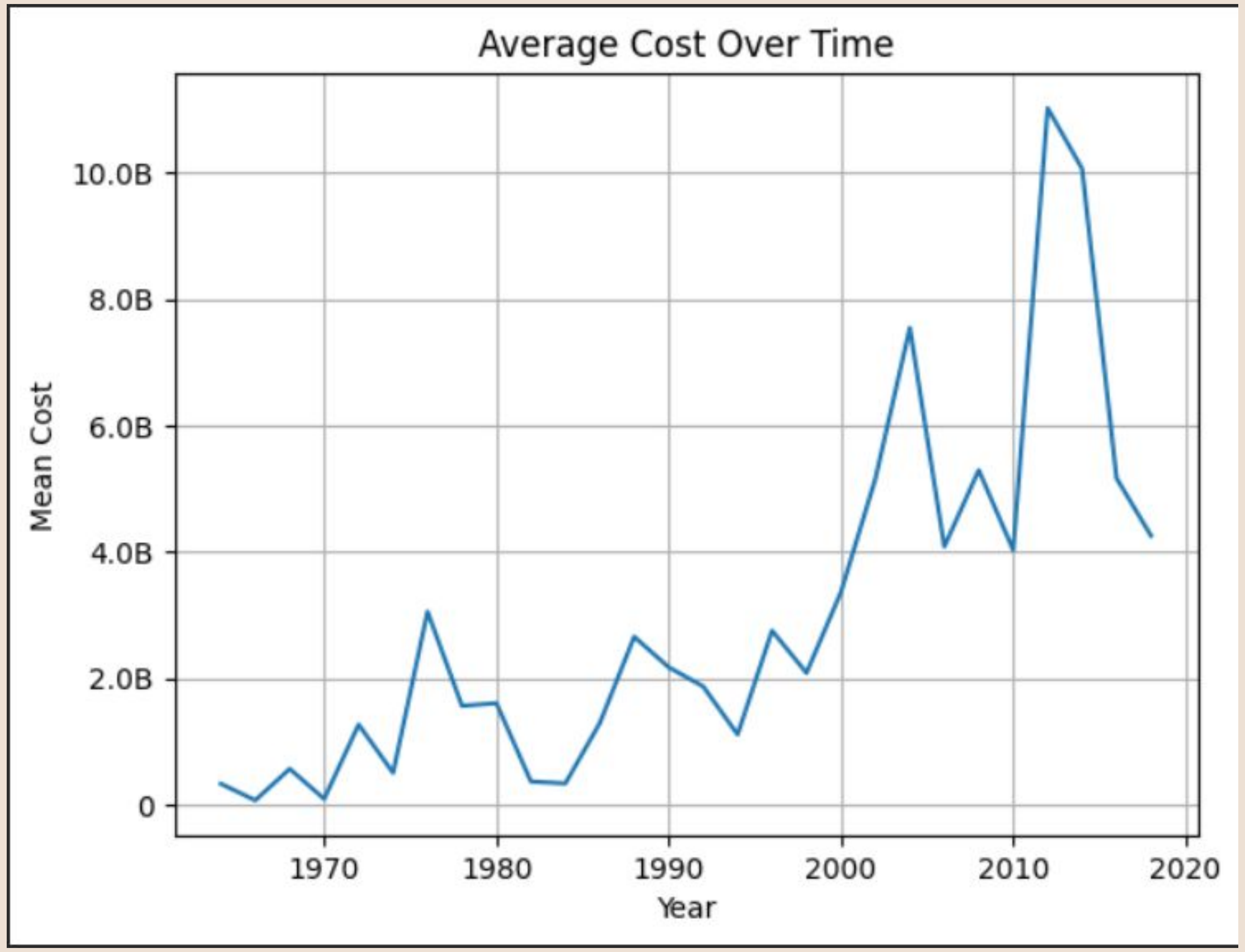
After cleaning, the dataset contains the following key numerical variables:

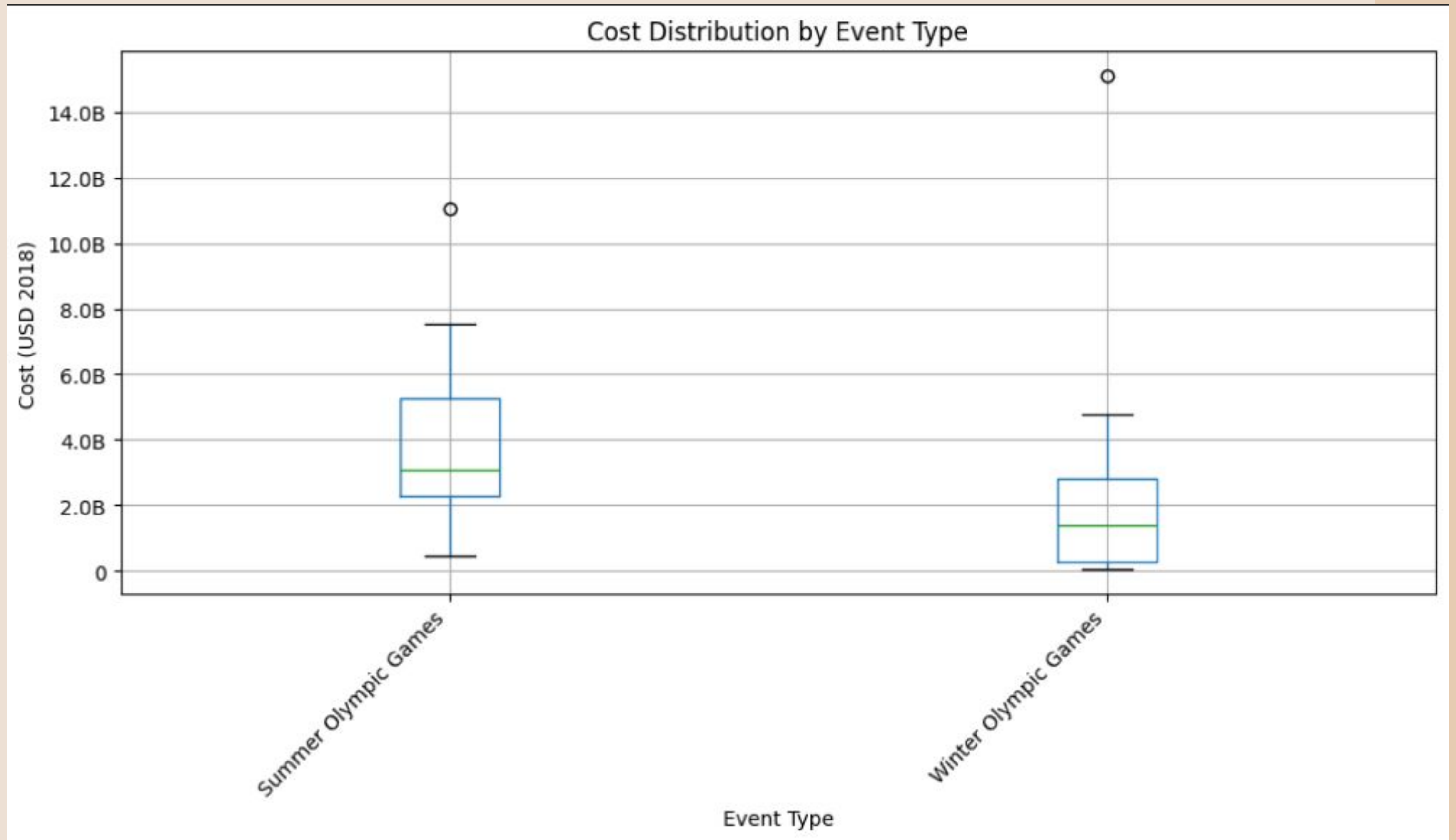
- Number of Events
- Number of Countries Represented
- Financial Metrics (USD 2018, inflation-adjusted): Different Revenue Types , Cost of Venues , Cost of Organisation , Total Cost (USD 2018) (aggregated from venue + organisation cost) , Total Revenue (USD 2018) (aggregated from ticketing + broadcast + sponsorships)

	Number of events	Number of countries	Ticketing Revenue (USD2018)	Broadcast Revenue (USD2018)	International Sponsorship Revenue	International Sponsorship Revenue (USD 2018)	Domestic Sponsorship Revenue (USD 2018)	Cost of venues (USD 2018)
count	29.0	29.0	29.0	29.0	24.0	29.0	0.0	29.0
mean	149.24	106.69	201.3M	892.9M	199.7M	232.5M	NaN	1.8B
std	102.18	60.31	254.6M	852.3M	211.4M	231.7M	NaN	2.7B
min	34.00	35.00	6.4M	7.3M	1.0M	846208.00	NaN	40.2M
25%	57.00	64.00	44.0M	108.1M	25.4M	28.3M	NaN	280.3M
50%	102.00	88.00	110.9M	693.7M	118.9M	157.4M	NaN	791.2M
75%	237.00	159.00	255.5M	1.4B	320.8M	359.2M	NaN	2.1B
max	306.00	207.00	1.1B	3.0B	668.0M	698.9M	NaN	12.6B

Exploratory Data Analysis







Model Training

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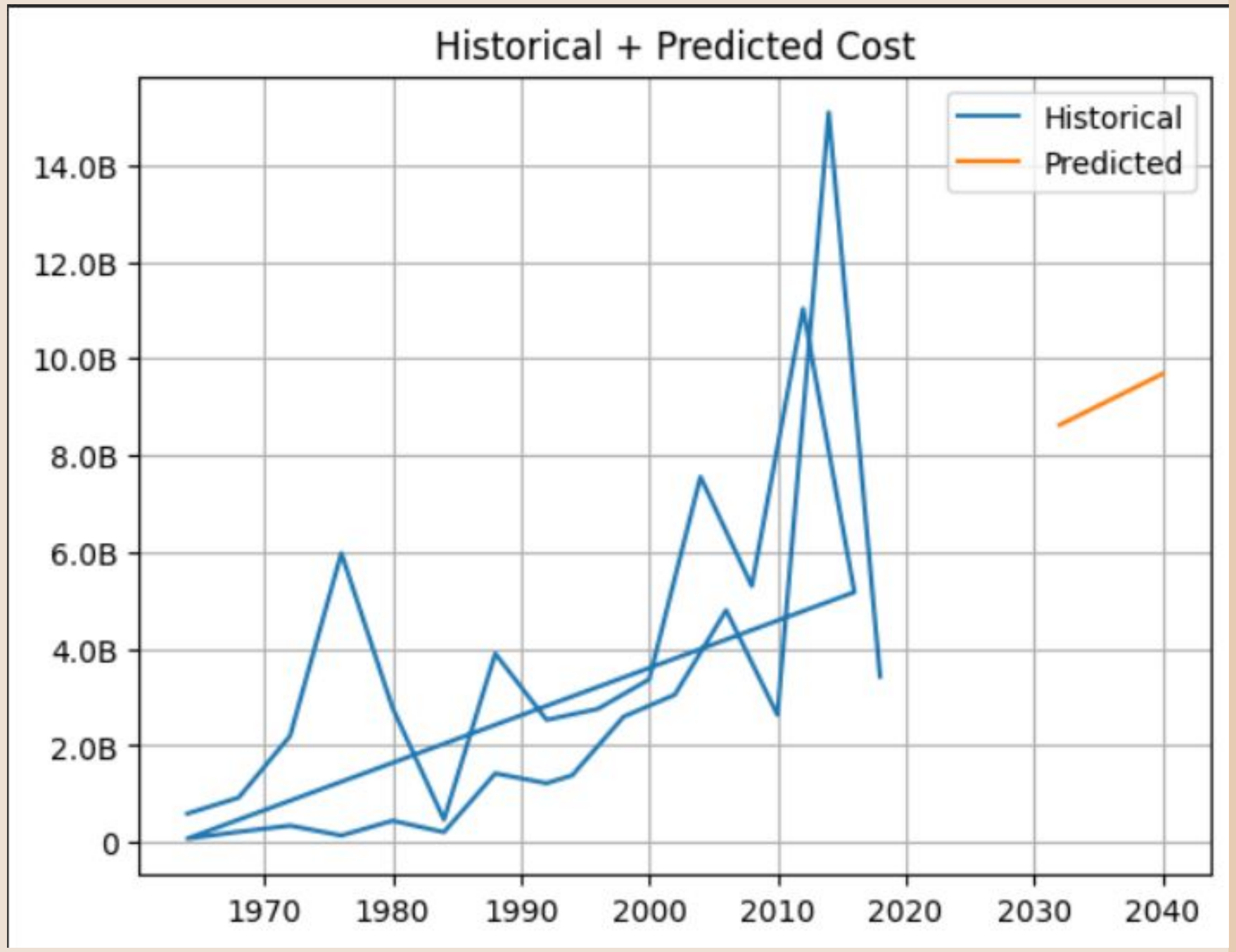
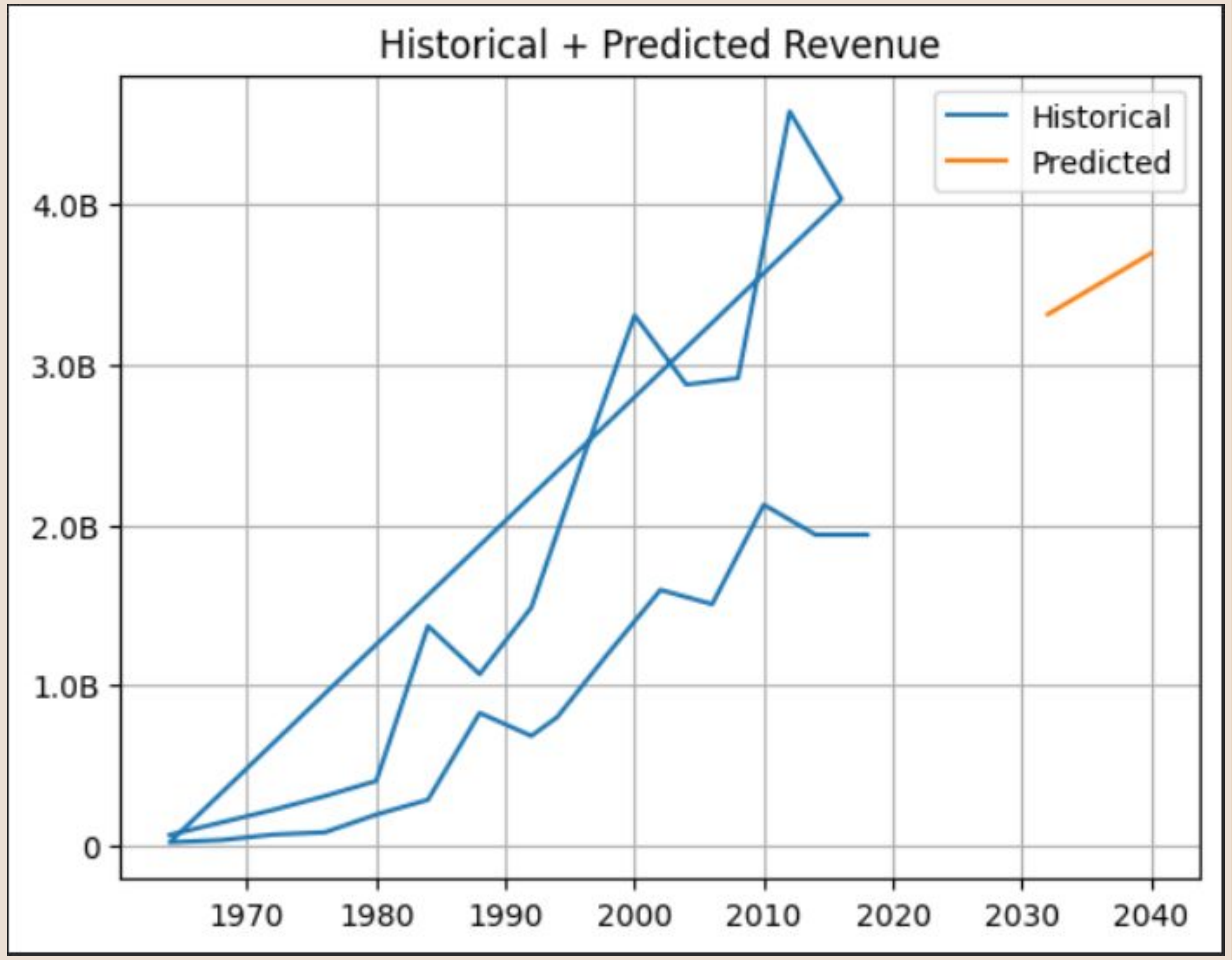
R2: -0.4116793887486512
MAE: 1263773788.641535
Slope: 0.3616527105685546
Intercept: 197409725.77811646
```

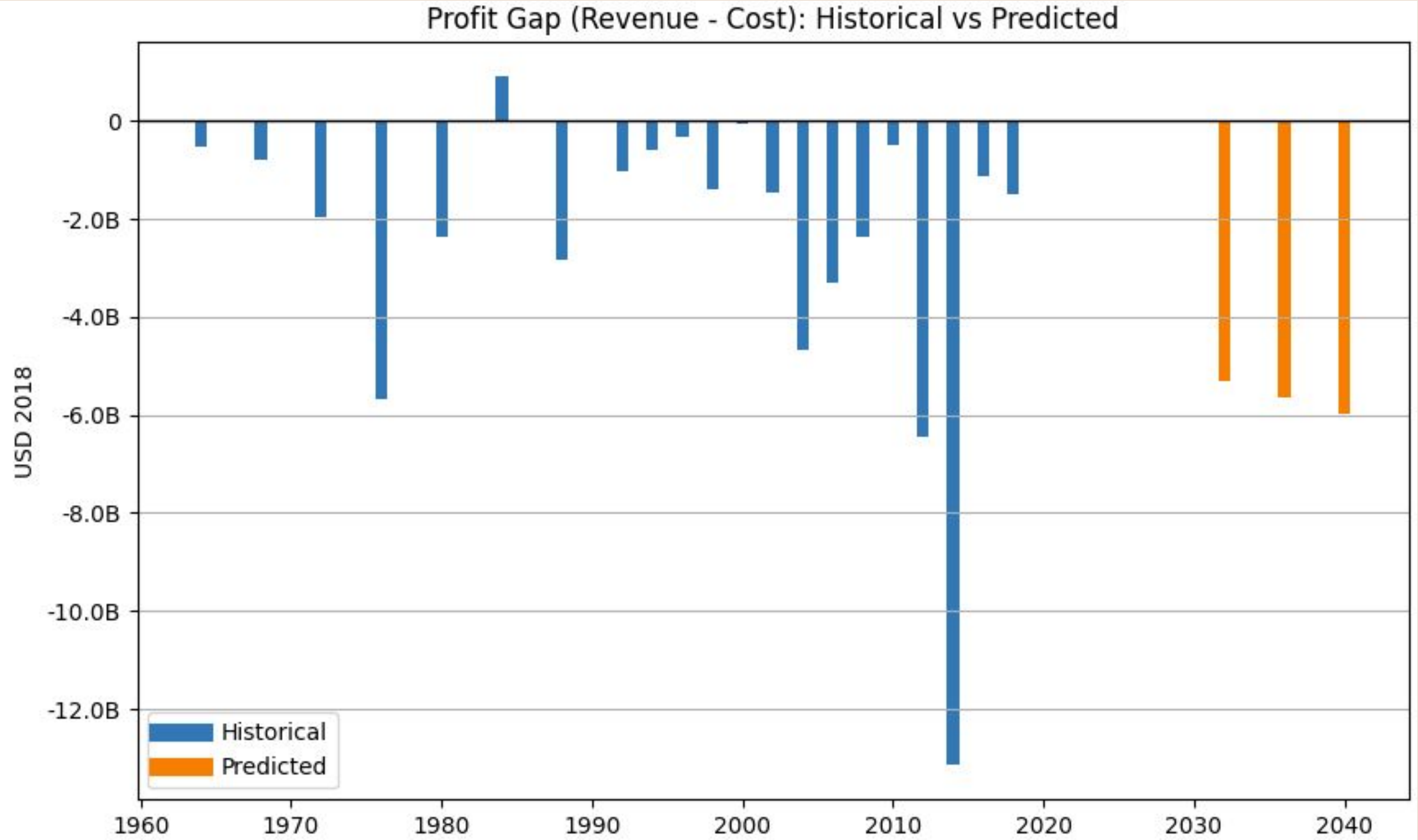
```
Slope: 131898988.49484275
Intercept: -259395441585.85007
```

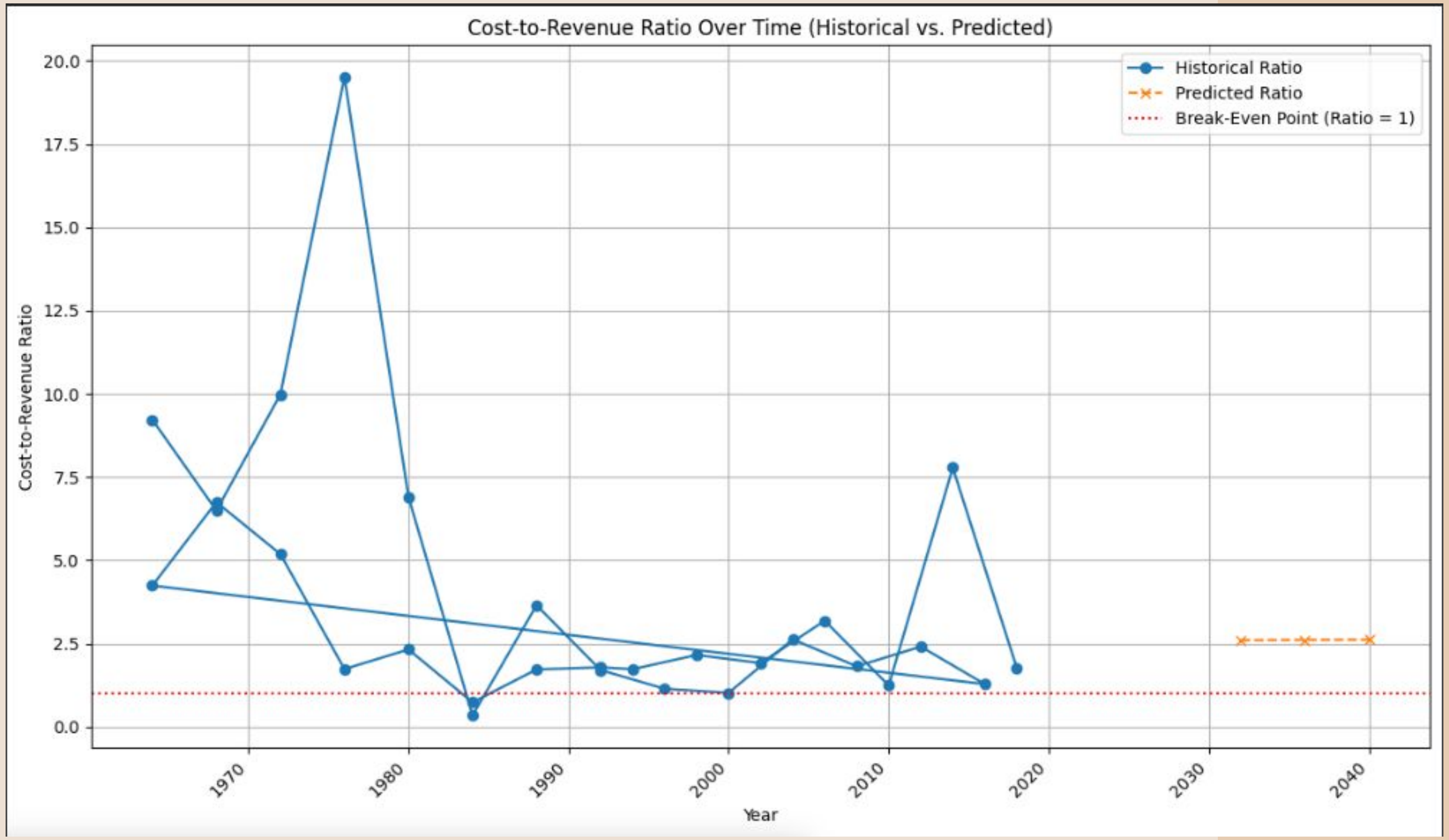
	Year	Predicted Cost (USD 2018)	Predicted Revenue (USD 2018)	Profit/Loss
0	2032	8.6B	3.3B	-5.3B
1	2036	9.2B	3.5B	-5.6B
2	2040	9.7B	3.7B	-6.0B





Analysis Result:





Random Forester Model:

```
Linear Regression Performance:
R2: -0.4116793887486512
MAE: 1263773788.641535

Random Forest Regressor Performance:
R2: 0.2342037269457573
MAE: 1040762171.5281504

Selected model for future predictions: RandomForestRegressor
```

- New predicted data

```
Year: 2032
  Predicted Cost: 6.0B
  Predicted Revenue: 1.8B
  Predicted Profit/Loss: -4.2B

Year: 2036
  Predicted Cost: 6.3B
  Predicted Revenue: 1.8B
  Predicted Profit/Loss: -4.5B

Year: 2040
  Predicted Cost: 6.6B
  Predicted Revenue: 1.5B
  Predicted Profit/Loss: -5.1B
```

Conclusion

- Historical analysis shows persistently negative profit gaps, with several years experiencing steep losses due to costs outweighing revenue.
- The cost-to-revenue ratio has rarely approached the break-even point (ratio = 1), indicating long-term financial inefficiency in the system.
- Predicted results maintain this pattern, suggesting that without major structural changes, losses are likely to continue in the future.
- The analysis highlights the need for improved cost controls, optimized operational strategies, and sustainable planning to shift toward profitability.
- Overall, the project provides data-driven insights into financial trends, helping identify where strategic improvements are most necessary.

Thank you!